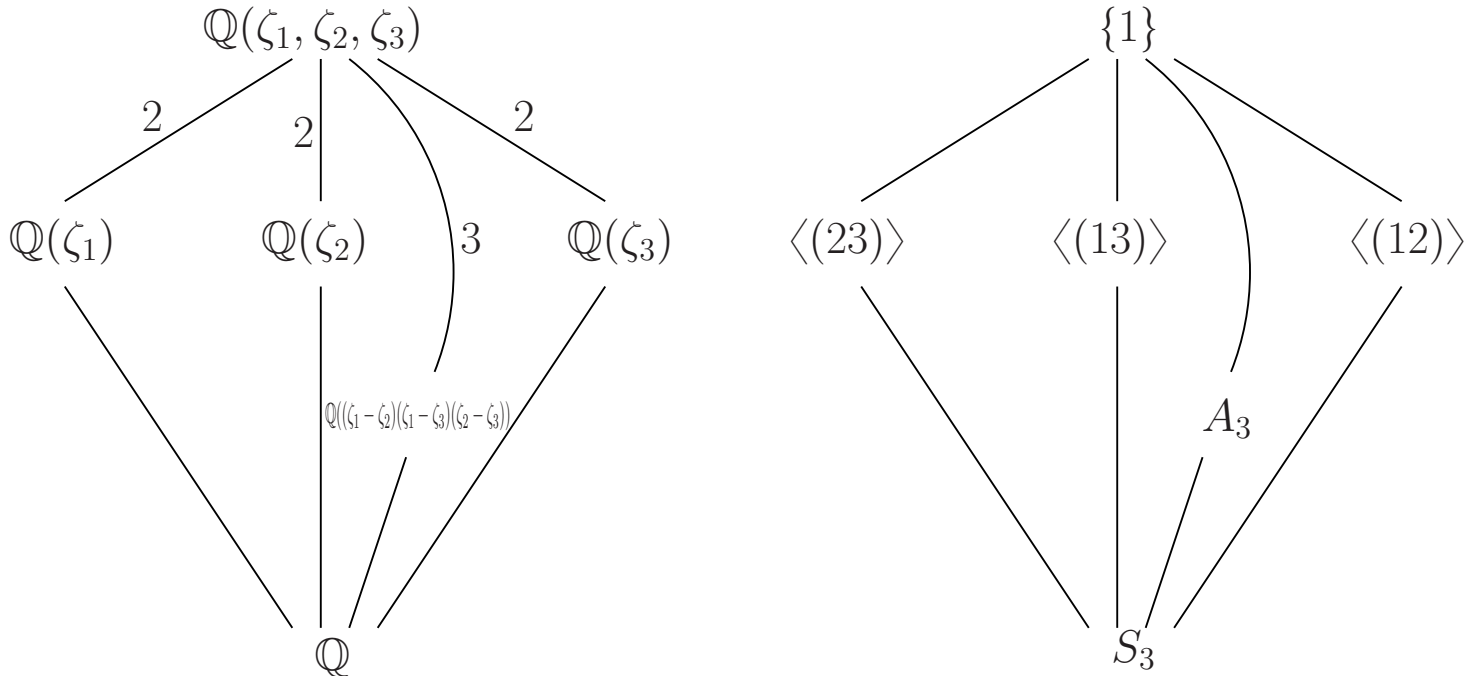


Note on Cubics with Largest Possible Galois Group

For any irreducible cubic $f \in \mathbb{Q}[x]$ with roots $\zeta_1, \zeta_2, \zeta_3 \in \mathbb{C}$ and splitting field L of degree 6 over $\mathbb{Q}$, the lattices of the subfields of L and the subgroups of $\text{Gal}_{\mathbb{Q}}(f)$ are respectively the following:



Note how the field extension degrees (denoted on the edges of the left-hand lattice) correspond exactly to the cardinalities of the corresponding subgroups of S_3 . The order-reversing correspondence between subfields of L and subgroups of S_3 follows easily from our observations today on the cubic discriminant. The one subtle part is deriving the subfield $K := \mathbb{Q}((\zeta_1 - \zeta_2)(\zeta_1 - \zeta_3)(\zeta_2 - \zeta_3))$ — the unique subfield K of L with $[L : K] = 3$ — from scratch, i.e., *without* using the Galois correspondence. We'll discuss this next lecture. It's worth noting that the corresponding lattices obtained by replacing $\mathbb{Q}$ with any field F of characteristic *other* than 2, and assuming $f \in F[x]$ is irreducible and has splitting field of degree 6 over K , remain the same.

Thanks to Hao, Joe, and Suleyman for catching my earlier error!